Company Bankruptcy Analysis

Purpose: perform exploratory data analysis (EDA) on a company bankruptcy dataset, identify the distribution of bankruptcy status, inspect data quality, compare financial indicators, and measure their association with bankruptcy using Spearman correlation.

1. Project Objective

The goal of the project is to analyze financial data for companies and investigate how selected financial indicators differ between bankrupt and non-bankrupt companies. The target variable is Bankrupt? where 0 represents a non-bankrupt company and 1 represents a bankrupt company.

2. Dataset Overview

Item	Result
Number of records	6,819 companies
Number of variables	96 columns
Target variable	Bankrupt?
Non-bankrupt companies	6,599 (96.77%)
Bankrupt companies	220 (3.23%)
Missing values	0
Duplicate rows	0

3. Libraries Used

Library	Purpose in the project
Pandas	Load the CSV file, inspect the DataFrame, group data, calculate statistics, and perform data analysis.
NumPy	Support numerical operations and provide numeric data types used in the analysis.
Matplotlib	Create and format charts.
Seaborn	Create statistical visualizations such as count plots and box plots.

4. Step-by-Step Work

4.1 Import the libraries

The required Python libraries were imported before starting the analysis.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

4.2 Load the dataset

The CSV dataset was loaded into a Pandas DataFrame named df.

```
df = pd.read_csv("data.csv")
```

4.3 Standardize column names

Leading and trailing spaces were removed from column names. This helps prevent errors such as KeyError when a column is referenced by name.

```
df.columns = df.columns.str.strip()
```

4.4 Inspect the dataset

These commands were used to understand the dataset size, view the first records, inspect data types and non-null counts, and calculate descriptive statistics.

```
df.shape
df.head()
df.info()
df.describe()
```

4.5 Check data quality

The analysis found zero missing values and zero duplicate rows.

```
df.isnull().sum().sum()
df.duplicated().sum()
```

4.6 Analyze the target variable

The number and percentage of bankrupt and non-bankrupt companies were calculated. This confirmed the strong class imbalance.

```
df["Bankrupt?"].value_counts()
df["Bankrupt?"].value_counts(normalize=True) * 100
```

4.7 Visualize bankruptcy distribution

A count plot was created to visually compare the sizes of the two target classes.

```
sns.countplot(x="Bankrupt?", data=df)
plt.title("Distribution of Bankrupt Companies")
plt.xlabel("Bankrupt? (0 = No, 1 = Yes)")
plt.ylabel("Number of Companies")
plt.show()
```

4.8 Compare ROA(C)

The mean and distribution of ROA(C) were compared between bankrupt and non-bankrupt companies.

```
roa_c = "ROA(C) before interest and depreciation before interest"
df.groupby("Bankrupt?")[roa_c].mean()

sns.boxplot(x="Bankrupt?", y=roa_c, data=df)
plt.show()
```

4.9 Compare ROA(A)

The same procedure was used for ROA(A) to compare its distribution across the two groups.

```
roa_a = "ROA(A) before interest and % after tax"
df.groupby("Bankrupt?")[roa_a].mean()

sns.boxplot(x="Bankrupt?", y=roa_a, data=df)
plt.show()
```

4.10 Calculate Spearman correlations

Spearman correlation was used to measure monotonic association between each numerical financial indicator and bankruptcy status.

```
spearman_corr = df.corr(method="spearman", numeric_only=True)["Bankrupt?"]
spearman_corr = spearman_corr.drop("Bankrupt?")
```

4.11 Select the top 10 indicators

The ten indicators with the strongest absolute Spearman correlations were selected. The signed values were retained so the direction of the association is still visible.

```
top10_signed = spearman_corr.loc[
    spearman_corr.abs().sort_values(ascending=False).head(10).index]
```

4.12 Select the top 5 indicators

The first five indicators were selected for a more detailed comparison using means and medians.

```
top5_features = top10_signed.index[:5].tolist()
```

4.13 Compare mean and median values

Both mean and median were compared because median is less sensitive to extreme values and therefore provides a useful complementary view.

```
mean_top5 = df.groupby("Bankrupt?")[top5_features].mean().T
median_top5 = df.groupby("Bankrupt?")[top5_features].median().T
```

4.14 Visualize the top five indicators

A grouped bar chart was created to compare the mean values of the five selected indicators across the two target classes.

```
mean_top5.plot(kind="bar", figsize=(12, 6))
plt.title("Mean of Top 5 Financial Indicators by Bankruptcy Status")
plt.show()
```

5. Main Results

5.1 Bankruptcy distribution

The dataset contains 6,819 companies. Of these, 6,599 are non-bankrupt (96.77%) and 220 are bankrupt (3.23%).

5.2 ROA(C) comparison

Status	Mean ROA(C)
Non-bankrupt (0)	0.508069
Bankrupt (1)	0.418503

The mean ROA(C) is lower for the bankrupt group in this dataset. The box plot also shows a lower central distribution for the bankrupt group, with several outlying observations.

5.3 ROA(A) comparison

Status	Mean ROA(A)
Non-bankrupt (0)	0.562015
Bankrupt (1)	0.456947

The mean ROA(A) is also lower for the bankrupt group in this dataset.

5.4 Top 10 indicators by absolute Spearman correlation

Feature	Spearman ρ
Persistent EPS in the Last Four Seasons	-0.238209
Net Income to Total Assets	-0.234316
Net profit before tax/Paid-in capital	-0.233106
Per Share Net profit before tax (Yuan ¥)	-0.232442
Retained Earnings to Total Assets	-0.229533
Continuous interest rate (after tax)	-0.227621
Total income/Total expense	-0.224564

ROA(B) before interest and depreciation after tax	-0.224163
ROA(C) before interest and depreciation before interest	-0.223589
ROA(A) before interest and % after tax	-0.222410

Interpretation: all ten selected indicators have negative Spearman correlations with Bankrupt?. A negative correlation means that higher values of the indicator tend to be associated with lower values of the bankruptcy-status variable in this dataset. Correlation does not prove causation.

5.5 Top five indicators: mean comparison

Indicator	Non-bankrupt (0)	Bankrupt (1)
Persistent EPS in the Last Four Seasons	0.230146	0.188818
Net Income to Total Assets	0.810083	0.738083
Net profit before tax/Paid-in capital	0.183883	0.147672
Per Share Net profit before tax (Yuan ¥)	0.185581	0.147765
Retained Earnings to Total Assets	0.935749	0.904244

5.6 Top five indicators: median comparison

Indicator	Non-bankrupt (0)	Bankrupt (1)
Persistent EPS in the Last Four Seasons	0.225111	0.195944
Net Income to Total Assets	0.811492	0.766297
Net profit before tax/Paid-in capital	0.179021	0.154012
Per Share Net profit before tax (Yuan ¥)	0.180236	0.155306
Retained Earnings to Total Assets	0.938058	0.912923

6. Conclusion

The exploratory data analysis showed that the dataset is large but strongly imbalanced toward non-bankrupt companies. The data contained no missing values and no duplicate rows. Several financial indicators showed negative Spearman correlations with bankruptcy status, and the five selected indicators had lower mean and median values for bankrupt companies than for non-bankrupt companies. These findings describe statistical associations in the dataset; they do not establish causation and do not determine the performance of a future